Hen House Automation (ESP32-S3)

An advanced, fully automated chicken coop controller built around the **ESP32-S3**, designed to manage the daily operation of a hen house with minimal intervention while prioritising reliability and safety.

Features

- 🐔 **Automatic coop door control** based on calculated sunrise and sunset times.
- 🌅 **Seasonal scheduling** with separate BST (summer) and GMT (winter) timing offsets.
- 💡 **Automated coop lighting** with configurable on/off offsets from sunset.
- 🌡️ **Environmental monitoring** using a DHT temperature and humidity sensor.
- 📺 **OLED status display** showing system status, environmental data, event times, Wi-Fi information, and diagnostics.
- 🌐 **Wi-Fi and NTP time synchronisation** with automatic RTC backup for accurate timekeeping.
- ⏰ **Real-Time Clock (RTC)** support to maintain operation during internet outages.
- 🔒 **Motor safety features** including limit switches, watchdog monitoring, fault detection, and automatic lockout protection.
- 🔔 **Status LEDs and buzzer alerts** for operating states, faults, and bedtime indications.
- 🔄 **Startup recovery** to safely restore the correct door position after power loss.
- 🎛️ **Manual override controls** for door and lighting with automatic timeout.
- 💾 **Persistent settings** stored in ESP32 Preferences memory.
- 📶 **Wi-Fi signal monitoring** and system diagnostics.
- 🛡️ **Designed for unattended 24/7 operation** with multiple layers of safety checks and recovery routines.

Hardware

- ESP32-S3 (N16R8)
- OLED display (U8g2 library)
- DHT temperature/humidity sensor
- DS3231/DS1307 RTC module (RTClib)
- Motor driver with open/close limit switches
- Status LEDs
- Piezo buzzer
- Manual control switches

Project Goal

The aim of this project is to provide a dependable, low-maintenance chicken coop automation system that opens and closes the coop door according to daylight hours, controls interior lighting, monitors environmental conditions, and continues operating reliably even during network or power interruptions. The firmware emphasises safety, fault tolerance, and long-term autonomous operation.